10. $\overset{+2}{Ba}\overset{-1}{Cl_2} + \overset{+1}{H_2}\overset{+6}{S}\overset{-2}{O_4} \rightarrow \overset{+2}{Ba}\overset{+6}{S}\overset{-2}{O_4} + 2\ \overset{+1}{H}\overset{-1}{Cl}$

11. $\overset{0}{P_4} + 3\,NaOH + 3\,H_2O \rightarrow \overset{-3}{P}H_3 + 3\,NaH_2\overset{+1}{P}O_2$

12. The reduction reaction of the oxidising agent(MnO_4^-) involves gain of 3 electrons. Hence the equivalent mass = (Molar mass of $KMnO_4$)/3 = 158.1/3 = 52.7

13. No. of moles of water present in 180 g

= Mass of water / Molar mass of water

= 180 g / 18 g mol^{-1} = 10 moles

One mole of water contains

= 6.022 x 10^{23} water molecules

10 mole of water contains = 6.022 x 10^{23} x 10 = 6.022 x 10^{24} water molecules

14. 7.5 g of gas occupies a volume of 5.6 liters at 273 K and 1 atm pressure Therefore, the mass of gas that occupies a volume of 22.4 liters

$$= \frac{7.5\,g}{5.6\,L} \times 22.4\,L = 30\,g$$

Molar mass of NO (14+16) = 30 g

15. No. of electrons present in one ammonia (NH_3) molecule (7 + 3) = 10

$$\text{No. of moles of ammonia} = \frac{\text{Mass}}{\text{Molar mass}}$$

$$= \frac{1.7\,g}{17\,gmol^{-1}}$$

$$= 0.1\,mol$$

No. of molecules present in 0.1 mol of ammonia

$$= 0.1 \times 6.022 \times 10^{23} = 6.022 \times 10^{22}$$

No. of electrons present in 0.1 mol of ammonia

$$= 10 \times 6.022 \times 10^{22} = 6.022 \times 10^{23}$$

16. $\overset{+3}{S_2}O_4^{2-} < \overset{+4}{S}O_3^{2-} < \overset{+5}{S_2}O_6^{2-} < \overset{+6}{S}O_4^{2-}$

17. $\overset{2+}{Fe}\ \overset{3+}{C_2}O_4 \xrightarrow{\text{Oxidising agent}} \overset{3+}{Fe} + \overset{+4}{C}O_2$

$n = 1 + 2(1) = 3$

11th Std Chemistry 229-289.indd 242 06-01-2021 16:04:27